Jamera Mellyn Fernando

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Education

University of California, San Diego Expected June 2028
B.S. Data Science, Minor in Cognitive Science *GPA: 3.6/4.0*
Relevant Coursework: Machine Learning, Data Management Systems, Data Structures & Algorithms, Scalable Analytics and Intro to Computer Systems
In Progress (Fall 2026): Deep Learning (PyTorch), Recommender Systems, Artificial Intelligence

Technical Skills

Languages: Python, SQL, JavaScript/TypeScript, C/C++, Bash/Shell
ML/AI: scikit-learn, XGBoost, LightGBM, Prophet, Word2Vec, PCA, LLM integration
Data & Cloud: PySpark, Dask, AWS (EMR, EC2, Lambda, S3), PostgreSQL, MongoDB, ETL pipelines
Software & Infrastructure: Git, Docker, Linux, CI/CD, REST APIs, automated regression testing

Experience

HDSI Lab 3.0 Fellowship La Jolla, CA
Data Science Lab Assistant *Oct 2025 – Present*
- Built **Python and C++ controls** for a fiber-optic display, integrating Raspberry Pi with Arduino motor-control nodes.
- Redesigned controller communication using **RS485**; validated operation across **20 motors**.
- Diagnosed **hardware and firmware failures**, resolving wiring, grounding, and communication issues through testing.

Village Up Remote
Project Manager *Jun 2026 – Jul 2026*
- Built an **AI pipeline** with OpenClaw and Telegram, automating application intake, resume parsing, and matching.
- Integrated Google Drive and Gmail APIs with OAuth; the system was spotlighted by Village Up and is piloting at San Diego State University with approximately **100 applicants**.
- Deployed and maintained the matching pipeline on a **Linux VPS**, using Git, SSH, and **automated regression tests**.

Out of the Blue Remote
Data Science Consultant *Apr 2025 – Apr 2026*
- Built **time-series anomaly detection** with Prophet, identifying **40+ critical incidents** across e-commerce metrics.
- Developed modular Python/SQL backend services powering **5+ KPI dashboards**.
- Preprocessed time-series data with Polars and Pandas, improving anomaly detection accuracy by approximately **20%**.

Rimble Berkeley, CA (Remote)
Data Engineering Intern *Jul 2025 – Aug 2025*
- Validated **S3 JSON ingestion** against structured schemas, resolving data quality mismatches with Python and boto3.
- Optimized **50+ AWS Lambda functions** supporting ETL workflows, reducing execution errors by approximately **30%** through systematic debugging and testing.

KlonIt Remote
Software Engineer Intern *Aug 2025 – Sep 2025*
- Built and shipped a cross-platform React Native frontend for an AI agent to **500+ early users**.
- Developed backend services integrating **LLM inference**, implementing prompt orchestration, response parsing, and error handling across the application's core pipeline.

Projects

Distributed Analytics Pipeline | Dask, AWS EC2, NumPy, Numba *Apr – May 2026*
- Processed **millions of records in ∼5 minutes** using Dask on AWS EC2, eliminating costly data shuffles.
- Benchmarked **cache locality, memory access patterns**, and matrix performance with NumPy and Numba.

ML Feature Engineering Pipeline | PySpark, AWS EMR, Word2Vec, PCA *Apr – May 2026*
- Built a **PySpark pipeline on AWS EMR** to extract nested fields and generate Word2Vec embeddings from products.
- Applied **PCA** to reduce feature dimensionality and trained a tuned Decision Tree Regressor on the engineered features.

Interactive Drought Visualization | Python, JavaScript, D3.js, CMIP6 *May – Jun 2026*
- Parsed, reshaped, and aggregated **CMIP6 climate data** to analyze regional drought patterns using SPI-12.
- Built an interactive data-storytelling website with geographic visualizations; presented findings to a class of **200+**.

Predictive Modeling for Food Ratings | Python, BeautifulSoup, scikit-learn *Nov 2025 – Dec 2025*
- Built a **scraping pipeline** to collect and structure food review datasets; engineered predictive features and ran EDA
- Trained and compared **three ML models** to analyze dataset bias and performance trade-offs.

Diabetes & High Cholesterol Risk Analysis | scikit-learn, XGBoost *Dec 2024 – Jan 2025*
- Analyzed **20,000+ NHANES records** to identify demographic risk factors; built a logistic regression baseline with **89.9% accuracy** and improved precision by approximately 10% with XGBoost.
- Presented model findings to peers, translating statistical results into clear insights about demographic risk factors.